



Spring 2026	Module Code:	Module Name: <i>Biomedical image informatics</i>
Course Code: <i>BIO441</i>		Maximum Marks: 100 Marks

Biomedical image informatics Lab Project

Project description:

Students will work in groups (maximum 6 students per group).
Each group will select a medical imaging dataset (e.g., X-ray, CT, MRI, Microscopy images).

The project must integrate:

1. **Classical image processing techniques studied in the course**
2. **Deep learning using Convolutional Neural Networks (CNNs)**

The goal is to investigate how classical image processing techniques can support and improve medical image classification using deep learning.

Stage 1 – Classical Image Processing & Feature Extraction

Each group must perform the following:

1. Dataset Exploration

- Load and visualize sample images.
- Display images using appropriate plotting methods.
- Generate and interpret histograms of pixel intensities.
- Normalize intensity values if needed.

2. Image Enhancement & Filtering

Apply at least:

- One smoothing filter (e.g., Gaussian or mean filter).
- One edge detection or feature detection method.

Display results before and after filtering.

3. Masking & Segmentation

- Create masks using thresholding or other segmentation techniques.
- Extract region(s) of interest (ROI).
- Display segmented objects clearly.

4. Feature Extraction

From the segmented region(s), extract:

- Mean intensity
- Area (number of pixels)

Stage 2 – Deep Learning & Comparison

1. CNN Model Development

- Implement a Convolutional Neural Network for classification.
- Clearly explain:
 - Architecture used
 - Number of layers
 - Activation functions
 - Loss function
 - Evaluation metrics

2. Training Experiments

Students must train and evaluate the CNN on:

- Raw images
- Preprocessed images (after applying segmentation or normalization)

3. Evaluation

For both experiments, report:

- Accuracy
- Precision
- Recall
- Confusion matrix

Note : Improvement is not required, but justification and interpretation are required.

Deliverables:

Each group must submit:

1. Source code (well-structured and commented)
2. A trained CNN model
3. A written report (PDF) including:
 - Problem description
 - Dataset description
 - Stage 1 results (with figures)

- Stage 2 results (with comparison tables)
- Discussion and conclusions

Each student must clearly state their contribution.

Date and discussion:

Practical Exam Week



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Lab Project Marking Criteria

Course Code:	BIO441	Course Name:	Biomedical image informatics	Assignment No.		Lab Project	Date:	-----												
Student Name:	-----					Student ID:	-----													
	A (89-100)				B (76-88)				C (67-75)				D (60-66)				F (0-59)			
	10	96	92	89	88	84	80	76	75	72	69	66	64	62	60	59	40	20	0	
Relevance & implementation of Function (40%)	All functions in program are relevant and correctly structured and implemented				Functions in program are correctly structured and implemented but there are few missing issues functions or minor deviations				Some Functions in program are not correctly structured and implemented. There are some missing issues or major deviations				Many functions are not correctly structured and implemented. There are many missing issues or major deviations				- Most of functions are incorrect for - There are major deviations in the implemented functions.			
Passing Test cases (40%)	The program can pass all test cases correctly.				The program can pass most of test cases correctly.				The program can pass only some of test cases correctly.				The program can pass only few of test cases correctly.				The program cannot run of the test or the cases syntax are such errors that the			
																	run at all.			

Clean Code (20%)	The written code follows the guidelines of clean code				The written code almost follows the guidelines of clean code				The written code partially follows the guidelines of clean code				Low ability to follow guidelines of clean code				The program does not follow the guidelines of clean code			
1 st marker Total							1 st marker Signature							ASU Agreed Mark						
2 nd Marker Total							2 nd marker Signature							UEL Agreed Mark						